Control work 4 · The circle

The Quarter IV assessment, and the four errors it produces every year.

LESSON 61 | 1 × 40 MIN | GEOMETRY 8, ТЕМЫ 61–62 | STAGE 9 · 7.1–7.3

LESSON CLOCK

2' | 36' | 2'

Setting up	2 min
The paper	36 min
Collect in	2 min

LEARNING OBJECTIVES

- Assess the circle chapter: tangents, chords, arcs and angles.
- Diagnose each lost mark by error type.
- Re-solve every task that was lost.

TEXTBOOK

National	Темы 61–62, pp. 149–151
Cambridge	Learner's Book pp. 142–162
Workbook	Workbook 7.1–7.3

01 Explanation

The paper (40 minutes)

Two variants of eight tasks. Tasks 1–5 are worth 1 mark, tasks 6–8 are worth 2:

- task 1 · classify a line from d and r
- task 2 · the tangent length from an external point
- task 3 · a chord from r and d
- task 4 · a central or inscribed angle
- task 5 · a cyclic quadrilateral
- task 6 · an angle from an internal or external vertex
- task 7 · an arc length or a sector area
- task 8 · a multi-step problem combining two rules

Marking note: in task 3 the half-chord must be doubled; a mark is lost if it is not.

The four errors

WORK ON MISTAKES

1. **The half-chord given as the chord** — an answer exactly half the right one.
2. **Inscribed and central angles confused** — a factor of two, in one direction or the other.

3. **Half the sum used where half the difference belongs** — the vertex was not located first.
4. **Arc length and sector area mixed up**, or the fraction $\frac{\alpha}{360°}$ forgotten entirely.

Each pupil writes the error number beside every lost mark and re-solves that task in full.

02 Worked examples

EXAMPLE 1 **Find the error: in a circle of radius *10*, a chord *6* from the centre has length *8***

1. $\sqrt{100 - 36} = 8$

   That is the **half**-chord.

2. The perpendicular from the centre bisects the chord.

   So the full chord is twice as long.

3. $a = 16$

ANSWER

16

EXAMPLE 2 **Find the error: an inscribed angle on a $80°$ arc is $160°$**

1. The rule was applied upside down.

   The **central** angle is twice the inscribed one.

2. $inscribed = \frac{1}{2} \cdot 80°$

3. $= 40°$

   And it can never exceed $180°$ anyway.

ANSWER

$40°$

03 Interactive model

Show the wrong working, take a vote on the error type, then reveal.

Diagnose the error

Claimed: $r = 13$, $d = 5$, so the chord is 12

1. $\sqrt{169 - 25} = 12$

Correct — but this is only half of the chord.

2. The perpendicular from the centre **bisects** the chord.

3. $a = 2 \cdot 12$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	OʻZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Work on mistakes	Xatolar ustida ishlash	Работа над ошибками
Tangent	Urinma	Касательная
Chord	Vatar	Хорда
Inscribed angle	Ichki chizilgan burchak	Вписанный угол
Arc	Yoy	Дуга
Half-chord	Yarim vatar	Половина хорды
Sector	Sektor	Сектор

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$r = 13$, $d = 5$. The chord is:

- 12
- 24
- 8
- 26

A central angle of $50°$ gives an inscribed angle of:

- $100°$
- $25°$
- $50°$
- $75°$

Vertex outside, arcs $120°$ and $40°$. The angle is:

- $80°$
- $40°$
- $160°$
- $60°$

A $90°$ sector of radius 8 has area:

- 4π
- 16π
- 64π
- 8π

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. ***Variant 1, task 1.*** *$r = 8$, $d = 11$. Classify the line.*

   ANSWER It misses the circle.

2. ***Task 2.*** *$OP = 10$, $r = 8$. Find the tangent length.*

   ANSWER 6

3. ***Task 3.*** *$r = 10$, $d = 8$. Find the chord.*

   ANSWER 12

4. ***Task 4.*** *A central angle is $120°$. Find the inscribed angle.*

   ANSWER $60°$

5. ***Task 5.*** *In a cyclic quadrilateral $\angle A = 96°$. Find $\angle C$.*

   ANSWER $84°$

6. ***Task 6.*** *Two chords meet inside, arcs $70°$ and $50°$. Find the angle.*

   ANSWER $60°$

7. ***Task 7.*** *Find the arc length of a $90°$ arc, radius 6.*

   ANSWER 3π

Medium · the standard question

7 PROBLEMS

1. ***Variant 2, task 1.*** *$r = 12$, $d = 12$. Classify the line.*

   ANSWER A tangent.

2. ***Task 2.*** *$OP = 17$, $r = 15$. Find the tangent length.*

   ANSWER 8

3. ***Task 3.*** *$r = 25$, $d = 7$. Find the chord.*

   ANSWER 48

4. ***Task 4.*** *An inscribed angle is $38°$. Find the arc and the central angle.*

   ANSWER $76°$ both

5. ***Task 6.*** *Two secants meet outside, arcs $130°$ and $50°$. Find the angle.*

   ANSWER $40°$

6. ***Task 7.*** *Find the sector area of a $120°$ sector, radius 9.*

   ANSWER 27π

7. ***Task 8.*** *AB is a diameter of a circle of radius 10 and $AC = 12$. Find BC.*

   ANSWER 16

Hard · stretch and reason

7 PROBLEMS

1. *A chord of 24 lies in a circle of radius 13. Find its distance from the centre and the central angle it subtends.*

   ANSWER $d = 5$; $\sin(\alpha/2) = \frac{12}{13}$, so $\alpha \approx 134.8°$

2. *Two tangents from P touch a circle of radius 8 with $\angle APB = 90°$. Find OP, PA and the area of OAPB.*

   ANSWER $OP = 8\sqrt{2}$, $PA = 8$, area 64

3. *A 60° sector of radius 12 is cut out. Find its arc, its area, and the perimeter of the sector.*

   ANSWER arc 4π, area 24π, perimeter $4\pi + 24 \approx 36.6$

4. *AB is a diameter of a circle of radius 17 and BC = 16. Find AC and the area of triangle ABC.*

   ANSWER $AC = 30$, area 240

5. *Two parallel chords of a circle of radius 10 are 12 and 16. Find both possible distances between them.*

   ANSWER $8 + 6 = 14$ or $8 - 6 = 2$

6. *A tangent–chord angle is 40°. Find the two arcs the chord cuts off.*

   ANSWER $80°$ and $280°$

7. *A circle of radius 10 has a chord subtending a 90° central angle. Find the area of the minor segment.*

   ANSWER $25\pi - 50 \approx 28.5$

07 Homework

Homework — 5 problems

Re-solve every task you lost marks on, with the error number written beside it.

1. Re-solve, in full, each task you lost a mark on.

2. Beside each, write the error number (1–4) that caused it.
3. *$r = 15$, $d = 9$. Find the chord.*
4. *Two secants meet outside, arcs 145° and 55°. Find the angle.*
5. *Find the arc length and the sector area of a 60° sector of radius 12.*